Highlights

- Real-time pipeline combining counterfactual scene graph attention with kinematic checks.
- CSGA scores six "what-if" manoeuvre hypotheses for every critical traffic agent.
- Bicycle-model validator (PIKV) rejected 46.0% of 678 hypotheses as infeasible.
- Counterfactual Risk Tensor drives a five-level graduated intervention policy.
- 372K-parameter model runs at 149.7 FPS (6.68 ms) on an RTX 3050 Laptop GPU.